Reviewed mixed-degree addition components

Cancellation: field and word

Input validity implies ordinate +/- alternatives, excludes mismatch branch; cancellation output has nonzero weights and normalized validity. Field; shared support; both normalized input divisibilities; four nonzero weights. Repeated root/zero ordinate allowed.

Exact prelude/degree-one output decode and canonicality; guards agree; conditional word cancellation validity. Prime 5<=p<2^31; 10 canonical coordinates; four nonzero weights; both normalized input validities.

Generic branch: field

Interpolation and explicit degree-two residual establish nonzero output weights and normalized validity under generic K!=0. Field; monic quintic without X4; four nonzero weights; both input validities; K!=0. Repeated output support and zero interpolation slope are allowed. This establishes polynomial divisibility, not a divisor-class sum.

Generic branch: literal UInt64

Six canonical decoded payloads; zero guard and Option correspondence; outer/generic K equality; conditional output validity. Prime 5<=p<2^31; 10 canonical coordinates plus curve[3]; validity adds nonzero weights, input validity and K!=0. The actual generic formula uses explicit squares; no pow loop. Literal output w=1 is canonical. The other curve coefficients enter the validity statement as field values.

What the reviews checked

Separate module, control and axiom readbacks passed. Cancellation review covers both field and word modules; generic field and generic word have separate reviews. Only standard Lean axioms were read back. Original failed attempts and control corrections remain in their packets.

Source assumptions and remaining branches

Machine scope is Field n=1, tables=None and normal counter execution. Pure value bundles may define unused outputs beyond an early-return guard; no operation-count or side-effect equivalence is asserted. No compiler/linkage, class-sum, tripling, secondary positive-coincidence output or whole-walk theorem is claimed. Polynomial controls do not assert smooth curve preprocessing.

Publication and software snapshot

Canonical integration is pending final root configuration. Boxed maintained Clippy: exit0, sources unchanged. Typed retained-data tests, full library and CLI checks remain pending root terminal review in this snapshot. These software gates are separate from mathematical component review.

Executed archival audit: fixed 420-case results

Coverage and dispositions

420 / 420 source identities in exact order; 60 targets, 7 cases each, selected by p=739 from a 180-record archive. Recomputed: 218 collector-ineligible; 202 eligible = 174 first occurrences + 28 redundant duplicates; 0 informative candidates. Native exit0. No missing, failed or post-terminal disposition.

The preserved pattern

All 28 duplicates are case6 versus case3 within the same target: designed -(G+Q) versus G+Q. Their canonically signed coefficient difference is (0,0). Across all60 targets the original case6 labels and divisors negate case3. Graph forwarding:174 edges,28 suppressions,0 cycles,0 terminal answers. These are coordinate-validation fixtures, not sampled collector paths.

Setup accounting: aggregate each target once

All60 ledger entries agree across their seven row copies.

Setup checks:3904 Jacobian operations;438198 M;8436 S;14252 I;760038 field additions.

Constructor:2640 Jacobian operations;299213 M;5631 S;9823 I;519770 field additions.

Reference, eligibility, adapter and graph phases remain separate; scalar and field counters are not conflated.

Independent review and exclusions

Reviewer independently recomputed coverage, split-root eligibility, signed normalization, duplicate classifications and the once-per-target ledger. Frozen source/control hashes matched. No native rerun: admission permitted one attempt. Native label checks verify D=aG+bQ and Q=kG against public known answers; controls include deliberately planted informative labels.

Hashing, comparisons, memory, primality arithmetic, dense reference arithmetic, internal expansion and complete runtime work are not fully priced. No comparative clock, natural-yield estimate, cost ratio or ranking is reported.

Scope retained

The original four-lead and five-followup criteria remain unchanged, including the full1,260-target requirement and separate18-slot upper gate. This snapshot adds component evidence and an archival correctness audit; previous aggregate snapshots are preserved. index.json binds exact producer and independent-review paths and hashes.